

5 Essential Data Engineering Skills

Discover the data engineering skills you need to thrive in the industry. Find out about the roles and responsibilities of a data engineer, and how you can develop your own skills.

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Updated Nov 22, 2023 · 11 min read



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Successful **data engineering** projects are heavily dependent on the data that's used for them. So the saying goes, garbage in, garbage out. Ensuring that data is collected, appropriately transformed, and made accessible to data scientists requires data engineering skills.

In this article, we will go over why data engineering is a good career choice in 2022, the four main groups that data engineer roles fall into, and the typical requirements to

get a job. If you're new to data engineering, we'll give you the top 5 skills you need to learn to get started in the field.

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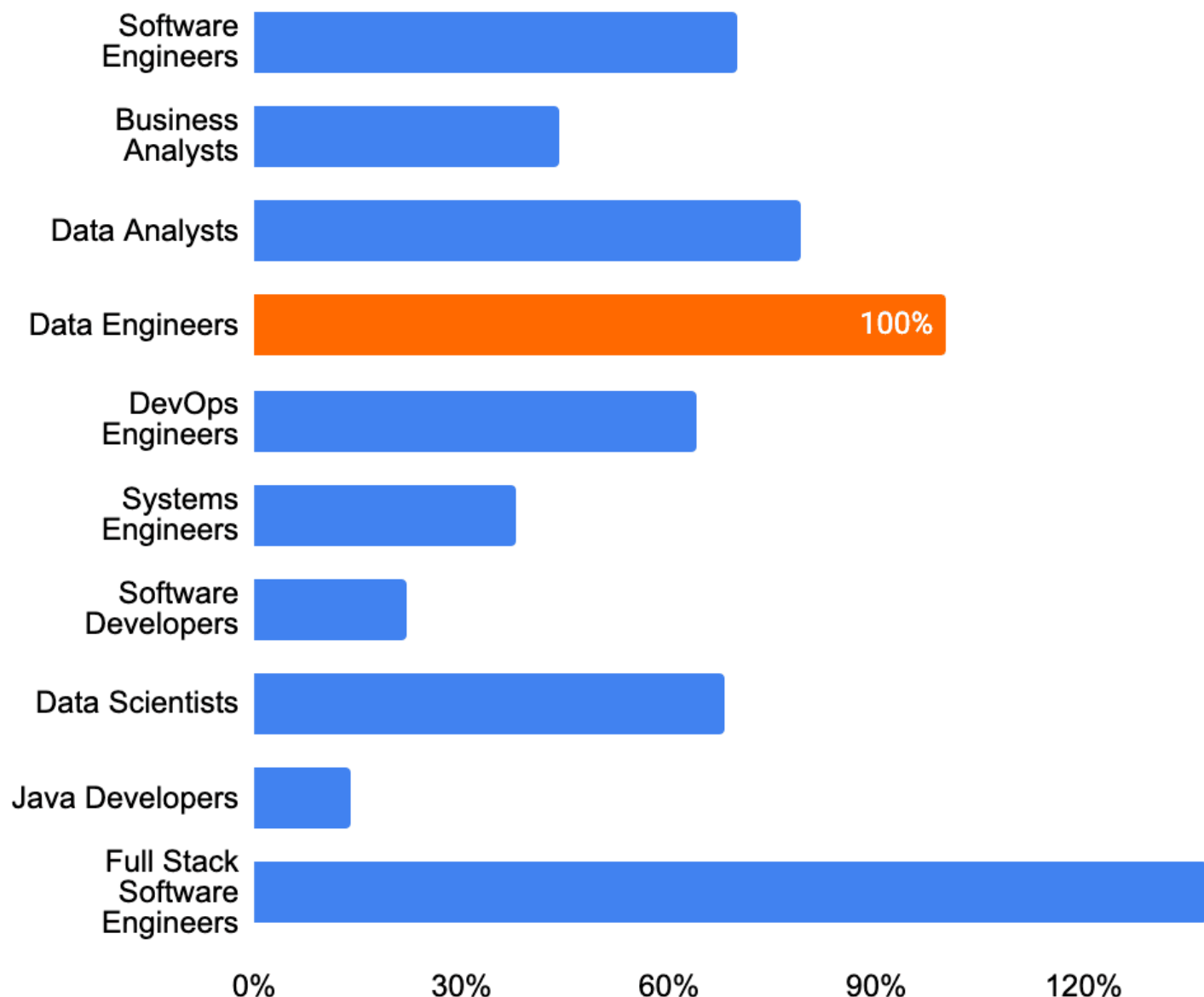
Why Pursue a Career in Data Engineering?

Almost 10 years ago, data science was declared the **sexiest job of the 21st century**. This lit a match under an already surging field, and data scientists started to explode onto the job market. However, along with the demand for analytics and predictive modeling, big tech giants like Facebook and AirBnB quickly recognized the need for the right people and tools to collect, store, manage and transform their data so that by the time it reached their data scientists, it is in a highly accessible state. Enter: **the data engineer**.

Data engineering has seen massive growth in the last couple of years. From 2021 to 2022, data engineering has grown by 100%, surpassing even that of the data scientist (68%). It also has the 4th highest volume of job postings compared to other tech roles. This shows the high demand for data engineers in today's job market.

The reality is that so long as data is used in a business to drive decision-making or answer business questions, the demand for data engineers will remain. So if you're interested in pursuing a career in data engineering, there has never been a better time.

YoY Growth of the Top 10 Jobs by Posting Volume



(Data source: [DICE](#), chart created by author)

Data Engineer Roles and Responsibilities

The [role of the data engineer](#) is extremely varied and entirely dependent on the size of the company and the technology and infrastructure they have. Companies with similar technology stacks can even hire data engineers for two completely different purposes.

That being said, the roles and responsibilities of data engineers typically fall into one of these four core groups:

- Generalists
- Specialists in data storage
- Specialists in programming and pipelines
- Specialists in analytics

Each one of these groups (except for the generalist) corresponds to a specific set of skills and tools that must be mastered to do your job effectively. Knowing which group you would like to work in can help to focus your learning efforts. Let's go over each of these groups.

Generalists

Data engineer generalists are involved in all aspects of data collection, storage, analysis, and movement. They are typically employed in small companies or companies in the early stages of analytics with small data teams.

The generalist is the hardest role in data engineering, especially for beginners. It can take many years of experience to learn and use the many different tools required by companies.

Specialists in Data Storage

Data engineers specializing in data storage are responsible for setting up and managing databases, data warehouses, and other storage platforms (both in the cloud and on-premise).

Some examples of data storage tools are:

- Relational and non-relational databases like SQL, NoSQL, and PostgreSQL
- Data warehouses like Redshift and Panoply
- Big data systems like Hadoop and Spark
- Cloud-based databases like AWS RDS and Microsoft Azure

These data engineers need a solid understanding of **data modeling techniques**. The chosen data storage platform should be optimized so that it operates effectively within the budget constraints of the company. Once a database or data warehouse is

designed and set up, it needs to be populated. An effective **ETL system** must also be designed to funnel in the data from possibly many different sources.

Specialists in Programming and Pipelines

Data engineers specializing in programming and pipelines are responsible for creating and managing the flow and movement of data. These data engineers must be familiar with many different programming languages and be able to integrate with many different platforms to **create data pipelines**, automate tasks, and write scripts.

These are the most common programming languages used by data engineers:

- Python
- Java
- C++
- Scala
- Ruby

Specialists in Analytics

Data engineers specializing in analytics work closely with data scientists and other analytics professionals. This means they must understand the tools, techniques, and frameworks used in data-related projects.

Depending on the project, data engineers must be familiar with many areas of data science and analytics, such as:

- Being able to set up and manage ETL tools and pipelines that support these projects (such as Stitch or Airflow)
- Being able to work with big data using tools like Hadoop, Spark, and Kafka
- Knowledge of BI tools and what they require, such as **Power BI** and Tableau
- Knowledge of machine learning libraries, such as Tensorflow, Spark, and PyTorch

Data Engineer Requirements

There are usually three main requirements that are considered for data engineer roles:

- Qualifications
- Certifications
- Experience

Most data engineers have either a bachelor's degree or some background in computer science, engineering, mathematics, or any other related IT field. The role of a data engineer requires a heavy amount of technical knowledge, which is why companies usually require at least a bachelor's degree. While it is also possible to get into data engineering without a technical degree, it is much more difficult, and you will need to do more to prove you have what it takes to do the job.

Certifications are good additions to your resume that can help set you apart from the competition. They prove that you have a good understanding of some of the frameworks or tools required for a job in data engineering.

Qualifications and certifications aside, it is often very difficult to get an entry-level position in data engineering. Companies typically ask for at least a few years of experience in a related field or using the required tools before considering a candidate.

This means you may need to use another data-related role as a bridge to get you into data engineering. It is common for someone to get hired at a company as a software engineer, business intelligence developer, or data analyst and then transfer to a data engineering role after gaining a few years of experience.

Top 5 Data Engineering Skills

Data engineering is an extremely broad and evolving field. There are so many tools, frameworks, and technologies out there that it is almost impossible to know and master all of them. The tools you choose to learn can depend on the company you want to interview for or which data engineer group you fall into.

However, for most data engineering roles, there are five crucial areas you need to develop. If you need somewhere to start, then start with these essential data engineering skills:

1. SQL Skills

SQL is not just a skill, but a cornerstone in the data engineering field. Mastery of SQL goes beyond basic queries; it involves understanding how to navigate and manipulate complex datasets using various SQL dialects such as NoSQL for unstructured data, PostgreSQL for object-relational database systems, and MySQL for its flexibility and reliability. To truly excel in data engineering, developing an in-depth knowledge of these variations is crucial.

If you're looking to get started with SQL, check out our [SQL Fundamentals track](#), which gives you a comprehensive introduction to Structured Query Language. You can also find our helpful [SQL Basics cheat sheet](#), which covers the essential queries you'll need to know.

2. Data Modeling Techniques

Effective data modeling is fundamental in data engineering, serving as the blueprint for building scalable and optimized databases and warehouses. It encompasses more than just the design; it involves understanding data relationships, constraints, and scalability. Mastering data modeling techniques is essential for executing efficient data pipelines, which form the backbone of data engineering projects, making this an essential data engineering skill.

You can get started with data modeling by using tools such as Power BI, and our course [Data Modeling in Power BI](#) is the ideal way to build your knowledge.

3. Python Skills

As far as programming languages go, Python is often considered as one of the most popular. With it, you can create data pipelines, integrations, automation, and clean and analyze data. It is also one of the most versatile languages and one of the best choices for learning first.

Python is so ubiquitous that many data engineering tools use the language in their back end and often allow for integration with data engineering tasks. To get started [learning Python](#), check out our [Data Engineer with Python track](#), which will teach you how to build an effective data architecture, streamline data processing, and maintain large-scale data systems.

4. Hadoop for Big Data Skills

Working with big data requires a specialized system, and Hadoop is among the most popular. It is a powerful, scalable, low-cost tool that has become synonymous with big data.

Organizations and individuals produce huge amounts of data on a daily basis, and data engineers will often have to maintain, test, analyze and evaluate these big data

[Get started with Hadoop](#) | [Big Data Engineer Interview Questions](#)



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The AWS cloud service is made up of services such as EC2, RDS, and Redshift. The use of cloud-based services has increased a lot over the years, and AWS is the most popular platform to get started with.

Data engineers need cloud computing skills, and you can start developing yours with our [AWS Cloud Concepts course](#).

To gain first hands-on experience with cloud data warehouses, try our [Exploring London's Travel Network Project](#). It offers a great opportunity to work with AWS Redshift, Google BigQuery, and Snowflake directly in your browser.

Bonus Data Engineering Skills: Soft Skills

Beyond technical prowess, soft skills play a pivotal role in a successful data engineering career. Problem-solving, teamwork, and effective communication with both technical and non-technical audiences are crucial. These skills enable data engineers to not only excel in their technical roles but also to collaborate effectively, innovate, and drive projects to success. They should be nurtured alongside technical skills for a well-rounded professional profile.

Final Thoughts

Data engineering is an increasingly vital field in today's data-driven world. Its significance is reflected in the diverse roles and responsibilities of data engineers, from generalists to specialists in storage, programming, pipelines, and analytics. The field offers a dynamic and challenging career path, with the opportunity to constantly learn and adapt to new technologies and methodologies.

For those considering a career in data engineering or looking to enhance their skills, the journey is both rewarding and demanding. Key skills like SQL, data modeling and Python, form the foundation of a competent data engineer's toolkit. However, remember that the technical prowess must be balanced with essential soft skills like problem-solving, teamwork, and effective communication.

If you want to pursue a career as a data engineer, our [Data Engineer with Python career track](#) will quickly get you up to speed on many of the core skills needed to get a job.

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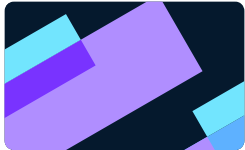
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